A Comparative Analysis of Machine Learning Models with Integrated Explainability for Automotive Fuel Efficiency Prediction

***Abstract*—Accurate forecasting of automotive fuel efficiency is essential for policy, consumer awareness, and engineering design, as a result of the increasing environmental and regulatory concerns that necessitate deeper understanding of fuel consumption. Nevertheless, the limited interpretability of predictive models and the complex relationships among multiple factors continue to present a challenge in the development of accurate prediction. This study proposes a complete machine learning framework for forecasting a vehicle's combined miles per gallon (MPG). We conducted a comprehensive evaluation of 12 cutting-edge algorithms, including linear models, tree-based ensembles, and support vector machines on a publicly available dataset (from Kaggle) of 550 automobiles from 2014 to 2024. Our methodology comprises unique domain-specific feature engineering and an integrated explainable AI (XAI) architecture that uses SHAP. The study found that Decision Tree had the highest prediction accuracy (R^2=0.9588) and Ridge had the best balance of performance and generalization (R^2=0.8469) on the test data. Crucially, our explainability study showed that the most important predictors were engineered features, specifically the displacement-year interaction and city-to-highway MPG ratio, providing useful information for automobile design. This study provides both a baseline for model selection and a clear, interpretable framework for fuel efficiency prediction.**

***Index Terms*—Fuel efficiency prediction, machine learning, explainable AI, feature engineering, automotive engineering, fuel consumption analysis**

I. Introduction

The automobile sector is quickly expanding, bringing a number of environmental issues, including rising fuel use and greenhouse gas emissions. Most cars run on petroleum-based fuels, which contribute to global warming, air pollution, and a number of health issues [1]. Most significantly, over 24% of the world's CO_2 emissions from fuel combustion come from the transportation sector, with passenger cars accounting for a significant share of this impact [2]. In addition to lowering fuel consumption, this environmental impact calls for the development of more effective fuel systems and improved vehicle performance. As a result, the building of sustainable transportation networks, customer decision-making, and regulatory compliance all depend heavily on accurate vehicle fuel efficiency prediction [3]. Standardized dynamometer tests and other traditional fuel economy methods are resource-intensive and frequently fall short of capturing the diversity of real-world driving circumstances [4]–[6].

Machine learning (ML) provides a possible alternative by discovering complex, nonlinear correlations between vehicle specifications and observed efficiency. The clustered structure of vehicle data, where various cars utilize different fuel types, is ignored by current studies [7]. From gradient boosting for conventional vehicles [8] to specialized models for electric vehicle range prediction, recent studies have shown the effectiveness of several ML techniques. Vehicle information is automatically grouped according to operating conditions, fuel type, and occupation. Data are assumed to be independent and identically distributed (IID) in standard ML, although this assumption is frequently broken. Models that were trained on mixed vehicle populations might not be able to generalize to new operating circumstances or unrecognized classifications. This clustering problem is often ignored in earlier research. By assessing 12 algorithms and examining their resilience across the intrinsic clusters in vehicle data, this study fills the gap.

Many of the models that are now in use in predicting fuel consumption, operate as "black boxes," missing the interpretability necessary for policy decisions and technical applications. It's just as important to comprehend the reasons behind a model's predictions as it is to know how accurate they are. Nevertheless, there aren't many thorough comparative studies in the field [9], [10] that assess a variety of cutting-edge algorithms and address the crucial requirement for model interpretability—a requirement for engineering applications where knowing why a model makes particular predictions is just as important as predictive accuracy itself.

By incorporating SHAP into a thorough comparison of 12 ML models—Linear Regression, Ridge, Lasso, Decision Tree, Random Forest, Gradient Boosting, AdaBoost, XGBoost, LightGBM, CatBoost, K-Nearest Neighbors (KNN), and Support Vector Regression (SVR)—on a secondary fuel consumption dataset from Kaggle, this study directly fills this gap. To the best of our knowledge, this thorough, explainable methodology has not been used to this particular dataset in any previous work. Four main contributions to the literature are made by our method:

- We assess twelve different ML algorithms, offering practitioners evidence-based recommendations for model selection based on generalization performance, accuracy, and computing efficiency.
- Moving beyond simple specifications to capture higher-order interactions, we create and evaluate ten unique features that encompass automotive engineering principles.
- We use SHAP, an XAI technique, to generate reliable and comprehensive insights into feature significance and model behavior.

- We translate model interpretations into actionable guidance for vehicle design by identifying key factors for improving fuel economy.

II. Related Work

The problem of fuel consumption forecasting has been widely studied, and traditional regression models have been proved to be not as effective in dealing with high-dimensional and dynamic information. Most recent developments in ML have greatly enhanced prediction accuracy. Nonetheless, the majority of ML models are black boxes, and they are not interpretable. It has resulted in the development of explainable artificial intelligence (XAI) that combines accuracy with transparency and trust on the side of the user [11].

Gradient Boosting Machine (GBM) and Support Vector Machine (SVM) were used to perform nonlinear regression on fuel consumption data when driving in real conditions, which indicates that the ML methods are superior to the conventional models because of their ability to capture complex nonlinear relationships [12]. Likewise, SVM is applied based on OBD-II data recorded an RMSE of 2.43, but the performance of SVM deteriorates with large datasets [13]. The Artificial Neural Networks (ANNs) have demonstrated certain potential, especially when engine speed and vehicle speed are used (models) compared to the Linear Regression and Random Forest [4]. The additional evidence that ANN can be used in predicting fuel consumption was the driving cycle data provided by GPS and CAN bus, although these data demand large preprocessing [14], [15]. More complex architectures such as RNNs, CNNs, and MLPs have since been developed, such that automatic extraction of features is possible. Extended contextual conditions that include the number of lanes and free-flow speed contributed greatly to the accuracy of predictions [16]. A backpropagation neural network with both time and space performed with 81.7% accuracy [17], but the working of ensemble methods to improve performance has been discussed in later studies, such as Random Forest, XGBoost, and LightGBM [18]–[20]. It has been empirically shown through comparative analysis of several ML models that tree-based ensembles and gradient boosting methods tend to be more accurate when carrying out fuel consumption prognostication tasks [3].

In more recent times, wavelet neural networks using stochastic gradient algorithms have performed better than the traditional ANNs in modeling nonlinear engine behavior [12]. Driving data aggregated in fixed-distance windows can be used to predict the fuel consumption of vehicles by vehicle-specific neural networks with high accuracy ($R^2 \approx 0.91$, with less than 4% error) [8], [15]. Although these developments include decision trees, CatBoost, linear regression, gradient boosting, Ridge, random forest, LightGBM, KNN, AdaBoost, XGBoost, Lasso, and SVR, most studies put more emphasis on accuracy at the expense of interpretability. The gap inspires the current study to focus on the integration of explainability methods with various ML models in the prediction of automotive fuel efficiency.

III. Methodology

In this section, we present the utilized dataset and its preprocessing, feature engineering steps, and dataset splitting. This is followed by model selection, training and hyperparameter settings, evaluation metrics, and explainability. The overall methodology is depicted in Fig. 1.

A. Dataset Description and Preprocessing

The dataset used in this study, comprises 550 car models from the 2014 to 2024 model years, sourced from Kaggle[1]. It focuses on essential performance metrics and specifications, making it suitable for analyzing fuel efficiency, engine configurations, and vehicle classifications across different manufacturers. It includes 11 features spanning vehicle class, model, manufacturer (make), engine specifications (cylinders, displacement), drivetrain (drive type, transmission), fuel type, and baseline efficiency metrics such as city and highway MPG. The target variable is the 'combination mpg', which represents the combined city and highway fuel efficiency. Table I presents a summary of the dataset description, including the feature name, a brief description, and the feature type.

TABLE I
Fuel Efficiency Dataset description

Feature	Description	Data type
combination_mpg	combined mpg	Numeric
city_mpg	city mpg	Numeric
class	vehicle class	Categorical
cylinders	engine cylinders	Numeric
displacement	engine displacement	Numeric
drive	type of drivetrain	Categorical
fuel_type	type of fuel	Categorical
highway_mpg	highway mpg	Numeric
make	manufacturer name	Categorical
model	car model name	Categorical
transmission	type of transmission	Categorical
year	manufacture year	Numeric

We applied the data preprocessing technique to standardize the fuel consumption analysis dataset prior to model training. The dataset underwent the following preprocessing procedures:

- Handling Missing Values: The median was used to impute the few missing entries for the variables 'cylinders' and 'displacement'.
- Handling Duplicate Values : Two duplicate records were found in the dataset and eliminated.
- Outlier Removal: A total of 10 outliers were detected and removed to ensure a standardized and robust dataset for model training.
- Encoding: Variables like 'drive', 'transmission', 'make', 'model', 'fuel_type', and 'class' were one-hot encoded.

[1]https://www.kaggle.com/datasets/arslaan5/explore-car-performance-fuel-efficiency-data

B. Feature Engineering

Feature engineering aims to improve ML models performance by converting raw data into meaningful representations, generating new features, and refining existing ones [21]. In this study, after a thorough analysis, certain features that were not significant or that could potentially cause overfitting were omitted, while ten additional features were included to capture essential automotive engineering and usage characteristics. The following ten features were added:

- `city_highway_ratio`: $\frac{\text{city_mpg}}{\text{highway_mpg}}$, characterizes a vehicle's efficiency profile across different driving cycles.
- `displacement_year_interaction`: displacement $\times$ year, models the evolution of engine technology and downsizing trends over time.
- `power_to_weight`: $\frac{\text{displacement}}{\text{cylinders} + 0.1}$, an approximation of specific power output.
- `car_age`: $2024 - \text{year}$, represents the age of the vehicle model relative to the analysis year.
- `displacement_per_cylinder`: $\frac{\text{displacement}}{\text{cylinders} + 0.1}$, average displacement per cylinder.
- `cylinders_displacement_interaction`: cylinders $\times$ displacement, captures the combined effect of engine configuration and size.
- `is_luxury`: binary indicator for luxury brands (audi, bmw, mercedes-benz, porsche, jaguar, land rover, genesis).
- `is_suv`: binary indicator for sport utility vehicle class.
- `is_4wd`: binary indicator for all-wheel drive or four-wheel drive.
- `is_automatic`: binary indicator for automatic transmission.

To prevent data leakage and overfitting, several features were excluded prior to modeling. The target variable `combination_mpg` was naturally removed. The raw efficiency metrics `city_mpg` and `highway_mpg` were dropped as they are direct components of the target and would artificially inflate performance. The `model` attribute, containing 276 unique values, was also excluded due to its high cardinality, which risks overfitting without contributing generalizable insights. After feature inclusion and exclusion, 18 features remained for model training as the final preprocessed dataset.

C. Dataset Splitting

The data was split 80 : 20 into training and test sets using the `train_test_split` function, with stratification based on the target variable to preserve distribution.

D. Model Selection, Training and Hyperparameter Setting

We implemented and trained 12 state-of-the-art models spanning major algorithmic families:

- Linear Models: Linear Regression, Ridge, Lasso.
- Tree-Based Ensembles: Random Forest, Gradient Boosting, AdaBoost, XGBoost, LightGBM, CatBoost, Decision Tree.
- Instance-Based & Others: k-Nearest Neighbors (KNN), Support Vector Regression (SVR).

All models were trained using a consistent pipeline with standardized 18 features. To achieve optimal performance, the hyper-parameters for a subset of models (Random Forest, Gradient Boosting, XGBoost, LightGBM, Ridge, SVR) were optimized via a combination of GridSearchCV and RandomizedSearchCV with 5-fold cross-validation, which was outlined in Table II.

TABLE II
OPTIMAL HYPERPARAMETERS SETTINGS IN OUR STUDY

Model	Best Parameters
RandomForest	n_estimators: 200, min_samples_split: 2, min_samples_leaf: 2, max_features: 'sqrt', max_depth: None
GradientBoosting	n_estimators: 200, min_samples_split: 2, min_samples_leaf: 1, max_depth: 3, learning_rate: 0.1
XGBoost	subsample: 0.8, n_estimators: 200, max_depth: 5, learning_rate: 0.1, colsample_bytree: 0.8
LightGBM	subsample: 0.8, num_leaves: 50, n_estimators: 100, max_depth: 7, learning_rate: 0.05
Ridge	alpha: 1.0
SVR	C: 10, epsilon: 0.1, kernel: 'linear'

E. Evaluation Metrics

Model performance was evaluated using the following metrics:

- R^2 (Coefficient of Determination): Measures the proportion of variance in the dependent variable explained by the model.
- RMSE (Root Mean Squared Error): Measures the average prediction error in MPG units.
- MAE (Mean Absolute Error): A more robust measure of average error.
- Training Time: Recorded to assess computational efficiency.

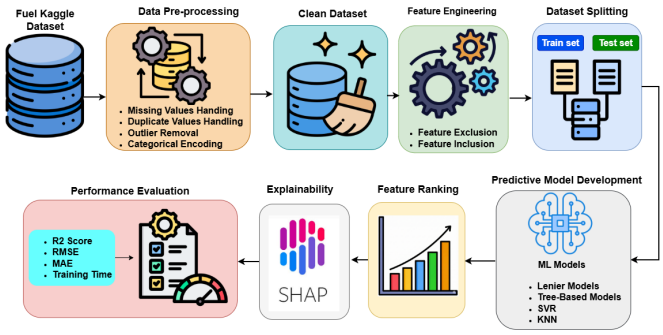

Fig. 1. Proposed architecture in predicting fuel consumption analysis

F. Explainability Framework

We used the SHAP (SHapley Additive exPlanations) framework [22] as an integrated explainability technique to improve transparency and interpretability. Our method combines predictive performance with transparency, bridging the gap between recent ML models that offer better accuracy at the expense of interpretability and classic regression approaches that struggle with high-dimensional and dynamic data. By quantifying each feature's marginal contribution to both individual predictions and overall model behavior, SHAP allows us to go beyond simple accuracy and comprehend the fundamental factors that influence fuel efficiency.

IV. Results and Analysis

A. Comprehensive Model Performance

Overall performance of the applied 12 state-of-the-art models is summarized in Table III. The Decision Tree achieved the highest test R^2 of 0.9588, closely followed by CatBoost (0.9571) and Linear Regression (0.9215). Notably, while tree-based models dominated the top rankings, the Ridge demonstrated balanced performance, an R^2 of 0.8469 with a smaller gap between training and test scores, suggesting better generalization. Moreover, the Decision Tree achieved the highest predictive accuracy, demonstrating minimal error in energy consumption estimation and effectively managing substantial data variability. Fig. 2 presents a comparison of model performance in terms of R^2 scores, where Decision Tree achieved the highest score, followed by CatBoost. Considering RMSE, the same set of models—particularly Decision Tree, recorded the lowest error of 2.026, falling below the 5.4 median mark, indicating superior predictive accuracy. In contrast, SVR exhibited the highest prediction error (RMSE) in this experiment.

TABLE III
Performance of Evaluated Models

Model	Train R^2	Test			Training (s)
		R^2	RMSE	MAE	
Decision Tree	**0.9999**	**0.9588**	2.026	1.091	0.01
CatBoost	0.9851	0.9571	2.069	1.196	0.69
Linear Regression	0.8821	0.9215	2.797	1.918	0.01
Gradient Boosting	0.9618	0.8858	3.374	1.573	0.12
Ridge	0.8371	0.8469	3.907	2.333	0.01
Random Forest	0.9381	0.8137	4.311	1.389	0.20
LightGBM	0.7246	0.5737	6.520	2.066	0.08
KNN	0.7591	0.5722	6.532	1.993	0.02
AdaBoost	0.8891	0.4775	7.219	2.619	0.04
XGBoost	0.9997	0.5015	7.051	1.544	0.13
Lasso	0.5541	0.4205	7.603	3.227	0.03
SVR	0.4368	0.1520	9.197	2.661	0.03

B. Impact of Hyperparameter Tuning

We performed hyperparameter tuning on the lower-performing models to evaluate its impact on predictive accuracy. The optimization yielded mixed results, as detailed in Table IV. While SVR showed dramatic improvement (+0.431 in R^2) and XGBoost improved substantially (+0.284), tuning provided negligible gains for Gradient Boosting and Ridge, and even degraded performance for LightGBM and Random Forest, which was also depicted in Fig. 3. This suggests that default parameters for some robust algorithms are near-optimal for this dataset, and excessive tuning can lead to overfitting to the validation fold.

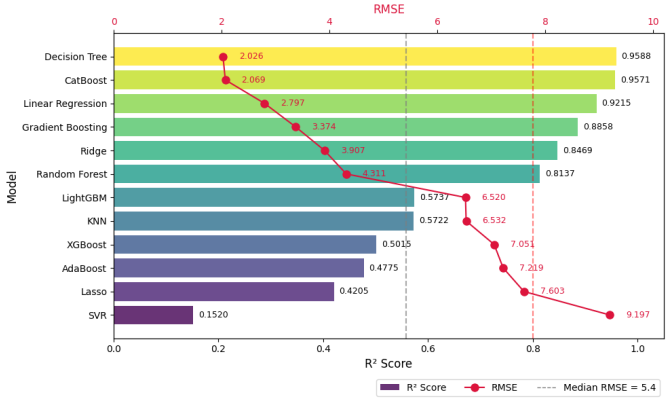

Fig. 2. Comparison of R^2 and RMSE scores across different models.

TABLE IV
Impact of Hyperparameter Tuning on Selected Models

Model	Original R^2	Tuned R^2	Δ R^2	Tuning Time (s)
SVR	0.1520	0.5827	+0.4307	2.72
XGBoost	0.5015	0.7858	+0.2843	2.44
Gradient Boosting	0.8858	0.8862	+0.0003	7.16
Ridge	0.8469	0.8469	0.0000	0.09
LightGBM	0.5737	0.4562	-0.1176	929.45
Random Forest	0.8137	0.6445	-0.1692	16.38

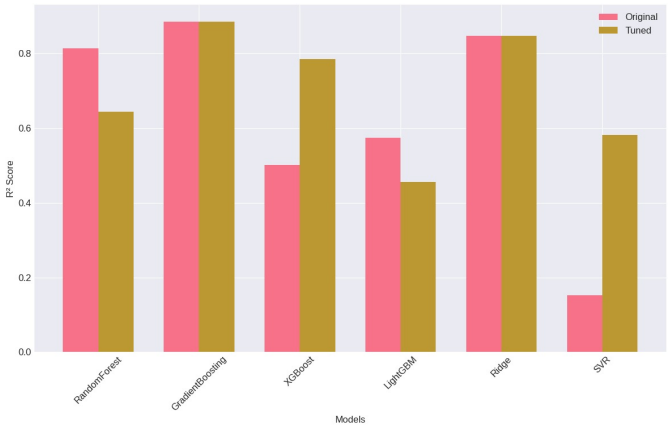

Fig. 3. Comparison of model performance (R^2 scores) between original hyperparameters and tuned hyperparameters, showing significant improvement for XGBoost and SVR after tuning, while RandomForest performance decreased.

C. Explainability Insights and Feature Ranking

SHAP was employed to measure the marginal contribution of each feature to both individual predictions and the overall model behavior. We used the Gradient-Boosting model to compute feature importance scores,

and visualized them using SHAP, as shown in Figure 4, the engineered feature `city_highway_ratio` emerged as the most influential predictor, followed closely by `displacement_year_interaction` and `drive_fwd`. This confirms that a vehicle's efficiency profile across driving cycles and the technological evolution captured through displacement-year interaction are the primary drivers of fuel economy. The presence of `fuel_type_electricity` among the top features further validates the significant impact of electrification on fuel efficiency.

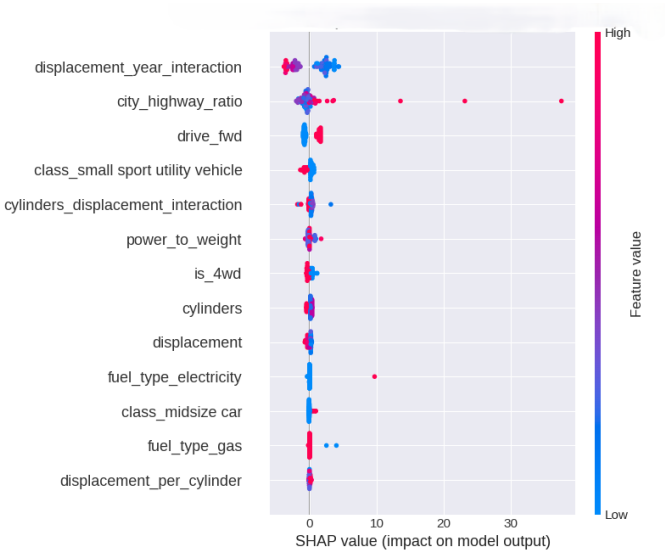

Fig. 4. Top 13 SHAP feature importance values, ranked by mean absolute SHAP value, from the GradientBoosting model.

While the SHAP explainability framework converged on a consistent hierarchy of feature importance. This indicates that a vehicle's efficiency profile (how it performs in city vs. highway driving) is more informative than its absolute city or highway mpg alone. The displacement_year_interaction and fuel_type_electricity were also highly important, highlighting the role of technological evolution and electrification, as evidenced by the Random Forest feature importance ranking of the top 15 features in Fig. 5.

V. Discussion

A. Synthesis of Key Findings

Several significant conclusions are drawn from our thorough investigation of fuel consumption prediction using a secondary dataset. For this structured, tabular dataset, tree-based ensemble techniques—Decision Tree and CatBoost in particular—provide the best prediction accuracy. Across a variety of performance evaluation parameters, Decision Tree demonstrated strong and consistent mpg prediction accuracy. Additionally, the Decision Tree model showed competitive and consistent prediction accuracy across all assessed variables. Their nearly perfect training results, however, point to a potential overfitting risk. As demonstrated by a lower train-

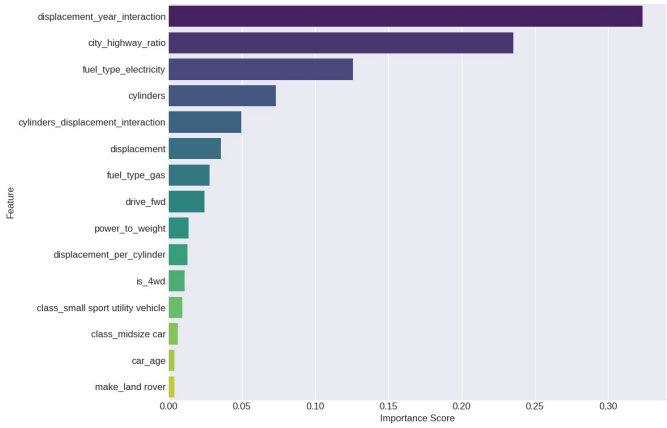

Fig. 5. The displacement-year interaction and the city or highway ratio are the most important predictors, according to feature importance scores obtained from the Random Forest model.

test gap, the Ridge offers a strong alternative with exceptional accuracy and much greater generalization.

The most important result is that the engineered 'city_highway_ratio' is extremely important. This highlights a crucial realization for the field: investing in the development of domain-informed features can pay off more than concentrating only on algorithm improvement. This is consistent with the more general ML idea that quality features are essential.

The trustworthiness of our feature importance conclusions is reinforced by the uniformity of results across SHAP. This method offers a more reliable basis for obtaining engineering insights and lessens the drawbacks of any one strategy.

B. Practical Implications for Automotive Engineering

Our models' interpretability directly translates into practical design and policy recommendations. The dominance of 'city_highway_ratio' indicates that concentrating only on highway aerodynamics may not have the same overall impact as increasing fuel efficiency in crowded, stop-start city driving (e.g., through improved start-stop systems, regenerative braking). Current industry trends are validated by the strong performance of 'displacement_year_interaction' and 'fuel_type_electricity'. Data-driven tactics to lower fleet-wide fuel usage include policies and research and development that support battery electric vehicles and smaller, turbocharged engines.

Gradient Boosting and Ridge are strongly suggested for a balance of speed, accuracy, and interpretability. Decision Tree or CatBoost are the best options for greatest accuracy when interpretability is not as important. Neural networks may be useful in the future for applications that need strong generalization on perhaps new future data.

C. Limitations and Future Work

The limitations of this study open up new avenues for investigation. Despite being useful, the dataset is small (n=550) and restricted to a few car features. Models could be further improved by extending the temporal range and adding more

precise technical details (such as vehicle weight, aerodynamic drag coefficient, and transmission gear ratios). A interesting next step is to investigate deep learning architectures created especially for tabular data, such as TabNet and FT-Transformer. Lastly, a useful way to increase real-world relevance would be to apply this paradigm to actual driving data gathered via telematics rather than standardized test cycles.

VI. Conclusion

This study offered a thorough, comprehensible ML framework for forecasting vehicle fuel economy. We found that instance-based ensembles provide the best generalization while tree-based ensembles, such as Decision Tree, reach peak accuracy through a methodical evaluation of 12 techniques. In terms of prediction accuracy and generalization capacity, ensemble tree-based algorithms like Random Forest and Gradient Boosting consistently beat linear models and individual base learners. Our most important discovery is that domain-specific feature engineering, namely the development of the 'city_highway_ratio', is a more potent lever for model performance than algorithm selection on its own. By clearly defining efficiency profiles, technical advancement, and electrification as the primary drivers of fuel economy, the integrated explainability framework turned prediction models into tools for insight generation. In addition to offering data scientists a useful standard, this effort offers automobile engineers and legislators working to improve transportation efficiency useful information.

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